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CO2-AT3- Class Test

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MLA0403

Deep Learning

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1. Differentiate between Machine learning and Deep learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.

Ans

Aspect	Machine Learning	Deep Learning
Architecture	use simple algorithm like Decision tree, SVM, random forest	uses Artificial Neural networks with multiple hidden layers
Data Requirements	works well for small-medium databases.	Requires very large databases for high accuracy.
Feature Extraction	manual feature engineering is required	Automatically learns features from raw data.
Computation	lower computation power	high Computational power (GPU/TPU)
Accuracy	Good for structured data	excellent for images, videos, speech and text.

Application of Machine learning

1. Email spam Detection
2. Credit card fraud Detection

Application of Deep Learning

1. Face Recognition
2. self-driving cars.

2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

Ans Representation Learning :-

Representation Learning is a technique in which a neural network automatically learns meaningful features from raw input data instead of relying on manually designed features.

Influence of width

- width refers to the no of neurons in each hidden layer.
- wider networks can learn more features immediately.
- Improves learning for simpler problems.
- Increases memory usage and computation.

Influence of Depth

- Depth refers to the no of hidden layers.
- Deeper networks learn hierarchical features.
- Better suited for complex tasks like image and speech recognition.
- may suffer from vanishing gradient and longer training time.

Comparison

Factors	width	Depth
Learning capability	Learns more features in one layer	Learns complex hierarchical features
Computational Complexity	more parameters per layer	more computations across layers
Model performance	Better for simple problems	Better for complex real-world problems.

3. Compare ReLU, Leaky ReLU (LReLU), and ELU activation functions explain their mathematical behaviour, advantages disadvantages and model performance.

Aspect	ReLU	LReLU	ELU
Mathematical formula	$f(x) = \max(0, x)$	$f(x) = x$ if $x > 0$ else αx	$f(x) = x$ if $x > 0$ else $\alpha(e^x - 1)$
Advantages	simple, fast, reduces vanishing gradient	prevents dead neurons	faster convergence mean close to zero
Disadvantages	Dead neuron problem	α must be selected	Computationally expensive
Applications	CNN, image classification	deep CNS	deep neural networks

Summary :

- ReLU : most widely used activation function.
- Leaky ReLU : Better than ReLU when neurons become inactive
- ELU : provides smoother learning and faster convergence

4. Explain the concept of unsupervised learning in neural network describe the working principle of Restricted Boltzmann machines (RBMs) and autoencoders and compare their objectives and applications.

Unsupervised learning :- Unsupervised learning is a learning method where the model is trained using unlabeled data. It discovers hidden patterns, structures and relationships without predefined outputs.

Restricted Boltzmann Machine (RBM)

- RBM contains of one visible layer and one hidden layer.
- There are no connections within the same layer.
- learns probability distribution of input data
- uses Gibbs Sampling and Contrastive divergence for training.

Application of RBM

- Recommendation systems
- feature learning
- Dimensionality Reduction.

Autoencoder :-

- Autoencoder consists of :
 - * encoder
 - * bottleneck
 - * decoder
- encoder compresses data
- decoder reconstructs the original input
- learns efficient representation by minimizing reconstruction error.

Application of ~~Auto~~ encoder.

- Image denoising
- Data Compression
- Anomaly detection
- Feature extraction

Feature	RBM	Autoencoders
Learning method	probabilistic	Neural Network Based
Objectives	learn probability distribution	Reconstruct input data
Architecture	visible + hidden layers	encoder + decoder
Applications	Recommendation systems	compression, denoising Anomaly Detection.

5. Explain the concept of Autoencoders and its uses in Deep Learning

Autoencoder

An autoencoder is an unsupervised neural network that learns to compress input data into a lower-dimensional representation and reconstruct it back with minimum loss.

Architecture →

1. Encoder
2. Latent space (Bottleneck)
3. Decoder

Working →

- Input data is passed to the encoder
- encoder generates compressed features
- Decoder reconstructs the original input.
- Training minimizes reconstruction error using loss functions such as mean squared error (MSE)

Uses of Autoencoder in DL:

1. Image Compression
2. Image Denoising
3. Anomaly Detection
4. Feature Extraction.
5. Dimensionality Reduction.

Advantages:

- learns feature automatically
- Reduce data

Disadvantages:

- requires sufficient training data
- may reconstruct noise if not properly trained.